Proof of Authorized Human Interaction for Voice AI Agents

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Abstract

AI agents ultimately derive their authority and accountability from the humans on whose behalf they act. As voice AI agents become capable of performing consequential actions, spoken prompts should be treated as authorization events rather than merely conversational inputs. Existing authentication mechanisms were designed to establish who entered a session, not who authorized each consequential command, leaving voice AI agents vulnerable to adversaries operating inside authenticated sessions through techniques such as injected audio, voice cloning, and stolen credentials. We argue that trustworthy voice AI agents require a new trust primitive: *Proof of Authorized Human Interaction (PAHI)*. PAHI introduces a new trust primitive that treats every consequential spoken prompt as a verifiable act of human authorization, progressively strengthening trust through interface-device authorization and physical-human authorization while preserving the low-friction nature of conversational interaction. We present PAHI as a multi-tiered foundation for building voice AI agents whose autonomy remains grounded in verifiable human authority.

CCS Concepts

• **Security and privacy → Hardware-based security protocols**; **Authentication**.

ACM Reference Format:
Isabella Lenz, Franziska Vollmer, Adil Ahmad, and Visar Berisha. 2026. Proof of Authorized Human Interaction for Voice AI Agents. In *ACM AI Summit 2026 (AI Summit '26), August 31-September 02, 2026, Atlanta, GA, USA.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3806096.3844842

1 Introduction

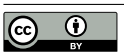
The modern capabilities of agentic AI necessitate a shift in the mechanisms we use to establish trustworthy authorization. Classic authentication guards the session boundary, assuming the adversary is an outsider and that once a human was authenticated, the actions that followed were theirs. Autonomous agents collapse that assumption: a single prompt can authorize a chain of consequential actions that unfold long after session authorization [9]. Within one authenticated session, an agent follows instructions from its user, other agents, and untrusted content *and cannot inherently distinguish among them.* An attacker holding the user's credentials can issue commands in the user's name, while prompt injection can steer the agent through untrusted content the user never authored. Account-level authentication can attest only that someone opened the session; it cannot vouch for any individual prompt issued within it [6, 17], let alone establish that the prompt came from a human at all. When the consequential action may be a single later prompt, the question is no longer who authenticated the session, but which human authored the command that caused the agent to act.

We currently do not posses a trust *primitive* for answering the aforementioned question. Most prior verification, from human perception to algorithmic biometrics, assumed that the media in question originated from a human. Voice and face biometrics therefore establish which person is speaking or present, not whether the content itself was human-generated. Liveness and anti-spoofing methods partially address this gap by detecting artifacts left by spoofing attacks [8, 14, 21]. Their effectiveness, however, depends on imperfections in synthetic media that continue to diminish as generative models improve [4, 20]. Moreover, they establish only that someone was momentarily present at a sensor, not that a particular command was authored by an authorized human. What has never existed is a durable, per-word attestation of human authorship, because until agentic AI it was never needed.

This need is particularly acute for voice-controlled AI agents. Voice is emerging as one of the most natural interfaces for directing autonomous systems, where spoken prompts increasingly authorize consequential actions [10, 15]. Rather than typing isolated requests, users increasingly direct agents through continuous spoken conversation, allowing them to execute tasks ranging from scheduling meetings and purchasing goods to controlling smart environments and financial transactions. As these systems move from answering questions to taking actions, spoken prompts become both conversational inputs and authorization events.

The baseline trust model for voice AI agents combines speaker verification at the prompt level with traditional account-level security [3, 7]. It assumes that the audio reaching the recognizer originated from the user's device rather than being injected by software, and that it was spoken by a live human rather than synthesized. A capable adversary operating inside an authenticated session can violate both assumptions.

We propose a layered authorization architecture in which each layer eliminates a distinct class of attacks, with stronger guarantees



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AI Summit '26, Atlanta, GA, USA
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ACM ISBN 979-8-4007-2638-5/2026/08
https://doi.org/10.1145/3806096.3844842

reserved for increasingly consequential actions. Tier 1 augments voice authorization with interface-device authorization: a hardware root of trust that attests that a command entered through a trusted microphone rather than being injected by software. Building on Tier 1, Tier 2 establishes physical-human authorization, providing evidence that the signal originated from a live human physically present at the device. The required level of authorization should scale with the action consequences: ordering lunch may require only Tier 1, but authorizing a wire transfer may require Tier 2.

The deepest tier addresses the one question that trusted hardware alone cannot answer. A secure microphone can attest that a signal entered through trusted hardware, but it cannot distinguish a spoken command from a synthetic one played into that same hardware [16]. A co-resident agent need only play a cloned version of the user's voice through the device's own speakers. What it cannot emit is a body.

We define physical-human authorization as proof of authorized human interaction (PAHI): a per-command attestation that a prompt was generated by a live, present human [13], sensed at the moment of utterance and bound through biometrics (e.g., speaker verification [3, 7]) to an authorized identity. Because the evidence is generated by the act of speaking itself, authorization occurs passively as part of the conversation. No purely digital adversary can satisfy this proof: copying every credential and secret a user possesses still does not place a physical body in front of the sensor. Physical-human authorization provides a new trust primitive for grounding high-consequence AI actions in verifiable human authority. The remainder of the paper defines the requirements for trusted voice authorization, presents the PAHI framework, demonstrates the feasibility of physical-human authorization through an mmWave-radar proof of concept, and concludes by outlining the grand challenges for operationalizing PAHI.

2 Requirements for Trusted Voice Authorization

As voice AI agents evolve from conversational assistants into autonomous actors capable of executing consequential tasks, spoken prompts become authorization events rather than mere inputs. Existing voice authorization mechanisms were designed to identify a speaker or secure an account, not to establish trustworthy authorization for every spoken command. We argue that any authorization mechanism for voice AI agents should satisfy three fundamental properties:

- **Per-prompt identity binding.** Every consequential prompt must be bound to the human who issued it. Existing mechanisms authenticate sessions, accounts, or devices, but they do not establish that an individual command originated from the enrolled user. Trust must therefore be established for each consequential prompt rather than assumed to persist throughout an authenticated session.
- **Low conversational friction.** Voice interfaces succeed because interaction is natural and continuous. Requiring passwords, passkeys, fingerprints, or other explicit confirmations for every command quickly becomes unusable. Authorization must therefore occur passively as part of normal conversation.

Table 1: Comparison of authorization mechanisms for voice AI agents.

Mechanism	Per-Prompt Identity Binding	Low Friction	Audit.
Session Authentication	No	Yes	No
Voice Authorization	Partial	Yes	Partial
Step-up Authentication	Yes	No	Partial
PAHI (Proposed)	**Yes**	**Yes**	**Yes**

- **Auditability.** Authorization should produce independently verifiable evidence linking agent actions to the human who authorized them. Without this, organizations cannot later demonstrate that a consequential action was genuinely authorized.

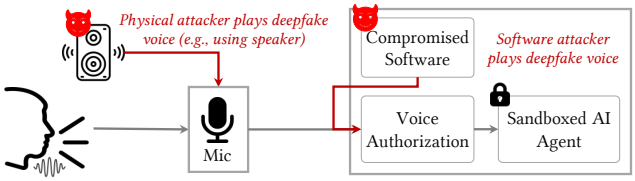
These three properties motivate our notion of *Proof of Authorized Human Interaction (PAHI)*, a per-command authorization mechanism that continuously binds spoken commands to an authorized human while operating transparently during natural conversation and producing durable evidence of authorization.

Table 1 compares representative authorization mechanisms for voice AI agents against the three requirements introduced above. Session authentication preserves low friction but neither binds identity to individual commands nor produces auditable evidence of authorization. Voice authorization improves identity binding, but cannot verify that a command originated from a trusted device or a live, authorized human, leaving both identity binding and auditability incomplete. At the other extreme, per-transaction step-up authentication provides strong identity binding and partial auditability, but only by interrupting the interaction with explicit user confirmation. PAHI is designed to satisfy all three requirements simultaneously by continuously binding each command to an authorized human, preserving the low-friction nature of voice interaction, and producing independently verifiable evidence of authorization.

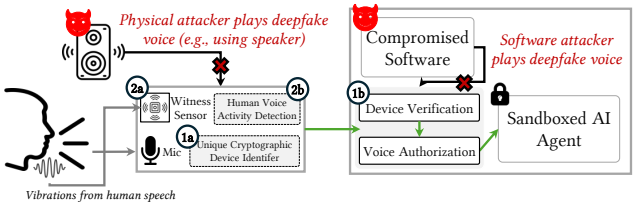
3 A Framework for PAHI

We envision the realization of PAHI in a *tiered and composable manner* for trustworthy authorization of AI agents. Figure 1**a** illustrates the voice authorization mechanism today where identity is established by verifying the voice sounds like an authorized human before an AI agent acts on the prompt. An attacker that operates within the same environment (e.g., malicious website script [5]) can easily break this by sending deepfake audio to the AI agent and invoke a privileged operation. Similarly, an attacker physically present can also play deepfake audio (e.g., using a speaker [16]) through the microphone to authorize activity. PAHI iteratively addresses these attacks (Figure 1**b**).

The first tier of PAHI continuously *establishes the identity of the human interface device from which a voice prompt originated.* This version of PAHI is suitable for enterprise and individual use-cases of AI agents where interface devices can be secured, and it prevents an attacker from injecting a synthetic voice prompt to authorize AI agents through software-only mechanisms and attacks. At the fundamental level, this requires adding an additional layer of unique device identity mechanisms (ⓘⓐ) to microphones that



(a) Current landscape of voice authorization for AI agents



(b) Envisioned landscape with two-tiered PAHI for AI agents

Figure 1: Current and envisioned landscape of voice authorization for AI agents.

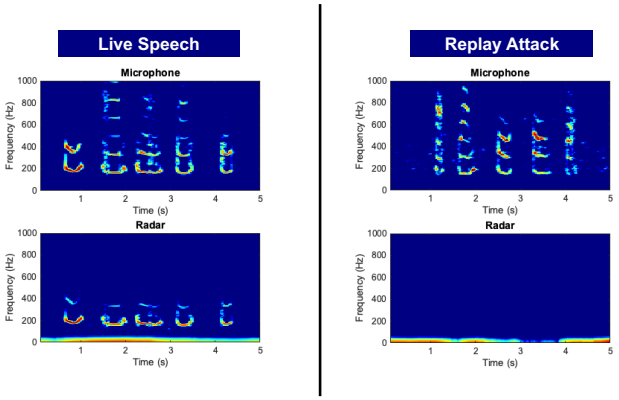


Figure 2: Microphone and radar spectrograms for live speech (left) and a replay attack (right). In both conditions, a person is physically present at the sensor, and low-frequency physiological activity is visible in the radar return. During live speech, the radar additionally captures neck-surface vibrations driven by vocal-fold motion that evolve synchronously with the acoustic signal recorded by the microphone. During the replay attack, the same audio is played through a loudspeaker held by the person; the microphone captures comparable acoustics, but the radar shows no vocal-fold vibration signature, only the cardio-respiratory low-frequency activity of the present individual. This contrast illustrates the core distinction PAHI exploits: the radar channel separates a live spoken utterance from a replay but by detecting the absence of biomechanical speech production signals.

can be used to sign outgoing audio packets to AI agents. Each authorized individual is provisioned their own unique device and identity chains are continuously established ((1b)) through end-to-end secure channels [1] with the sandboxed agent.

In highly security-sensitive contexts (e.g., government and military applications), it may not be sufficient to establish only that a command entered through a trusted device. PAHI therefore composes into a second tier that extends voice prompt authorization from trusted device identity to *the physical origin of the prompt, providing evidence that it was produced by the biomechanics of an authorized human.* Speech is produced through the coordinated action of multiple physiological systems. During phonation, air from the lungs passes through the vocal folds in the larynx, causing them to vibrate rapidly. These vibrations generate acoustic pressure waves at a fundamental frequency typically between 80–240 Hz in adults [12]. We propose to attach a secondary *Witness Sensor* (e.g. a millimeter-wave (mmWave) radar) to capture complementary physical measurements of the audio producing event from an independent sensor ((2a)). By binding these two modalities physically and temporally to a single event ((2b)), witness sensing provides evidence that the recorded content originates from something that actually occurred in the physical world [13].

The aforementioned tiers satisfy our three requirements described in the previous section. In particular, identity binding is established in the first tier through unique (and authorized) device ownership, while the second tier extends that to unique human signals. From a human's perspective, our envisioned setup operates within current workflows (i.e., human still speaks normally) and does not add a new layer of conversational friction to authorization. Finally, each authorization event can be traced back to the interface device from which it originated from in the first tier, and additionally the unique human it originated from in the second tier, ensuring auditability.

4 Radar as a Voice Witness Sensor

Tier 2 of PAHI requires an independent sensor that can provide evidence that a spoken command was physically produced by a live human. To demonstrate that this is feasible, we instantiate Tier 2 using mmWave radar as a witness sensor for the biomechanical activity underlying speech production [11, 13]. This section demonstrates a proof of concept for how this works.

The microphone records the acoustic waveform, while the radar measures neck-surface vibrations generated during speech together with cardio-respiratory activity of the human speaking. Figure 2 contrasts live speech with a replay attack. During live speech, the radar and microphone capture synchronized manifestations of the same speech event and the low-frequency cardio-respiratory signal of the speaker. During replay, the microphone records the reproduced audio, but the radar contains no corresponding speech-production signature; however the cardio-respiratory signal is still captured as there is a human in front of the microphone.

These radar and microphone signals provide two forms of evidence. A cross-modal model verifies that the radar and microphone measurements originate from the same physical speech event, establishing evidence of authorship. This is evident in the figure by comparing the high-frequency portion of the radar spectrogram with the acoustic spectrogram. Independently, the radar detects physiological activity associated with a living speaker, establishing evidence of liveness. Together, they provide the physical evidence

required for Tier 2 PAHI. A more comprehensive offline evaluation of the cross-modal verification methods and liveness algorithms, including quantitative results across multiple experimental conditions, is presented in [11].

Unlike conventional liveness checks, which establish only that a person is present at a sensor, witness sensing links the complete spoken command to the human biomechanics that produced it. Replaying cloned or synthesized speech reproduces the acoustics but not the underlying vocal-fold motion or physiological activity, forcing an attacker from the digital domain into the physical one.

The proof of concept demonstrates the central premise of Tier 2 PAHI: an independent witness sensor can provide physical evidence that a spoken authorization originated from a live human. Realizing PAHI requires extending this capability to continuous, real-time, secure authorization.

5 Grand Challenges on the Path to PAHI

Realizing PAHI as a deployable authorization primitive is a multidisciplinary challenge spanning signal processing and biometrics, machine learning, embedded systems, cryptography, and standards and governance. These disciplines must transform the physical evidence demonstrated in the preceding section into an authorization mechanism that is robust, deployable, and trustworthy.

The challenges differ across the two tiers of PAHI. Tier 1 (interface-device authorization) can largely be assembled from existing technologies such as the Device Identity Composition Engine (DICE) [18] and the Trusted Platform Module (TPM) [2, 19]. Its remaining challenges are primarily engineering: integrating these building blocks into compact, low-power communication devices. Tier 2 (physical-human authorization) is less mature. While the preceding section demonstrates the feasibility of witness sensing, substantial research is still required to transform it into dependable, continuous, real-time authorization.

Other challenges cut across both tiers. Beyond sensing, PAHI requires standards for device certification, credential revocation, and authorization auditing. Because authorization ultimately governs what an AI agent may do on a person's behalf, human factors are equally important: authorization must remain proportional to consequence while remaining robust to coercion and deception. We organize remaining research agenda around open problems below.

Presence, authorship, and identity. Presence, authorship, and identity are distinct guarantees that are easily conflated. Point-in-time liveness establishes only that a person is present at the device. PAHI instead requires evidence that the utterance itself was physically produced by that person. This demands a witness modality with sufficient temporal resolution to align the biomechanics of speech production with the acoustic signal throughout the utterance. Identity is a separate question: which enrolled individual spoke. Witness sensing can establish authorship and contribute complementary biometric features, but robustly fusing these with voice biometrics (or biometrics from other modalities) across aging, distance, posture, context of use, and spoofing remains an open problem.

Real-world validation. Initial validation of witness sensing has been obtained under controlled offline conditions with cooperative speakers and under replay conditions [11]. PAHI must instead operate continuously during natural conversation, in noisy and uncontrolled environments, across diverse populations, and in benign and adversarial conditions. The open problem is robustness and demographic generalization under spontaneous speech and motion achieved *passively*, since explicit challenge prompts would violate the low-friction requirement that makes voice authorization usable. In addition, resilience to adversarial spoofing through mechanical replay attacks (e.g. loudspeaker playback) and time-aligned manipulation of genuine witness traces must be evaluated.

Attestation on the edge. The witness sensor observes involuntary physiological signals and bystanders, making its raw data among the most sensitive a system could collect. Authorization must therefore be computed entirely on the edge. Sensing, cross-modal verification, and liveness decisions should execute on-device, producing only a compact, cryptographically verifiable authorization claim while transmitting no raw witness data. Achieving this within the power and compute constraints of edge devices is essential for making PAHI both practical and trustworthy.

Consequence calibration and authorization. PAHI requires a principled way to determine which authorization tier an action warrants, since the same spoken command may have very different consequences depending on context. It also requires an explicit authorization step that binds consequence, liveness, authorship, and identity to a specific command before releasing an authorization credential that is independently verifiable and revocable. A separate human-factors challenge remains.

Physical-human authorization establishes presence and authorship, but not willingness. Users may be coerced, deceived, or overheard. Designing authorization that remains robust to these threats with low-friction remains an open problem.

Interoperability and trust infrastructure. For PAHI to function as a trust primitive, authorization claims must be portable across devices, vendors, AI platforms, and services. This requires standardized formats for authorization credentials, interoperable verification protocols, and governance for device certification, credential revocation, and auditing. Just as cryptographic certificates became useful only through a shared trust infrastructure, PAHI will require standards that allow independent systems to verify and rely on physical-human authorization.

6 Conclusion

This work outlines a vision for a two-tiered architecture that grounds every action taken by a voice agent to an authorized human interaction either at a trusted device-level or the physical world. The latter is envisioned to be achieved using radar hardware that witnesses the physical event of speech production.

Ethics and Privacy Statement

PAHI is designed to be privacy-preserving: it computes on the edge, records no semantic content, and retains no raw witness signal. However we recognize residual risks: the sensing can capture involuntary physiology and bystanders, collected biometric traces could be repurposed, and reliability may vary across populations.

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